



Instant Checkout Credit Engine

A Multi-Modal Risk Inference System for BNPL & Microloans
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Scope

1. Problem motivation
2. Proposed technical solution
3. Expected outcome
4. Project timeline
5. Potential challenges



The Problem

Credit Exclusion: Traditional FICO scores turn away legitimate users who simply lack standard banking history.

Latency Drops: Standard loan assessments take too long. E-commerce checkouts require sub-second decisions to prevent cart abandonment.

Data Waste: Valuable alternative unstructured data (e.g., transaction notes, merchant logs) is completely ignored by legacy tabular systems.



Proposed Technical Solution

Tabular Track

XGBoost Engine

Processes structured financial numbers (transaction velocity, debt ratio). Highly optimized using Optuna for rapid numerical execution.

NLP Track

DistilBERT Model

Extracts dense semantic embeddings from unstructured transaction text to capture alternative credit signals from unbanked users.

Fusion API

FastAPI & Docker

A late-fusion classification head merges both streams via Dockerized asynchronous APIs to guarantee real-time, low-latency responses.

Expected Outcome



<100ms

Inference Latency

Instant Lending API

A production-ready, containerized dual-model API capable of serving real-time credit decisions without interrupting the user's checkout flow.

Includes SHAP explainability to satisfy strict financial regulations by outputting explicit feature attributions for every approval or rejection.



Project Timeline

Weeks 1-3

Architecture design,
synthetic data generation,
environment setup, and
Firebase backend
initialization.

Weeks 9-10

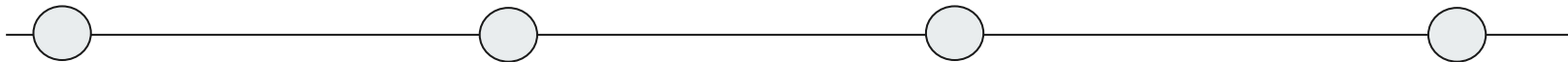
Late-fusion head engineering,
FastAPI endpoint integration,
and SHAP explainability
setup.

Weeks 4-8

Parallel Track Dev:
Tabular team engineers
XGBoost pipeline. NLP
team handles DistilBERT
fine-tuning.

Weeks 11-12

End-to-end testing, Docker
deployment, latency
optimization, stress testing,
and documentation.





Potential Challenges

Inference Latency

Risk: Transformer models (DistilBERT) are computationally heavy, threatening our strict 100ms target.

Mitigation: We will distill the NLP model size heavily and aggressively cache common token embeddings (e.g., frequent merchant names) in memory.

Model Explainability

Risk: Financial regulations strictly require transparent reasoning for credit rejections. Black-box networks fail this.

Mitigation: Integrating SHAP/LIME frameworks to output explicit, readable feature attributions alongside every decision to ensure compliance.



Thank you!